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Authors

Dr. FADHIL HIDAYAT
Corresponding Author
Submitting Author

Affiliations

- School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Dr. KUSPRASAPTA MUTIJARSA

Affiliations

- School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Mr. ARI APRIANSYAH

 [ORCID](https://orcid.org/0009-0005-9709-1504)
<https://orcid.org/0009-0005-9709-1504>

Affiliations

- School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Dr. ARRY AKHMAD ARMAN

Affiliations

- School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Dr. I GUSTI BAGUS BASKARA NUGRAHA

Affiliations

- School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Prof. SUHONO HARSO SUPANGKAT

Affiliations

- School of Electrical Engineering and Informatics,
Bandung Institute of Technology, Bandung 40132,
Indonesia

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Half-Flip 2D for Robust Facial Emotion Recognition in Engagement Evaluation during Online Learning

FADHIL HIDAYAT¹, KUSPRASAPTA MUTIJARSA¹, ARI APRIANSYAH¹, ARRY AKHMAD ARMAN, I GUSTI BAGUS BASKARA NUGRAHA, AND SUHONO HARSO SUPANGKAT

School of Electrical Engineering and Informatics, Bandung Institute of Technology, Bandung 40132, Indonesia

Corresponding author: Fadhil Hidayat (fadhil_hidayat@itb.ac.id)

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ABSTRACT Engagement evaluation in online learning relies heavily on facial emotion recognition (FER) to capture learner attentiveness, affective involvement, and cognitive focus. However, FER performance is often disrupted when students partially cover their faces with hands or adopt non-frontal poses, conditions frequently observed during instructional sessions. Such occlusions and pose variations reduce recognition reliability and weaken engagement assessment. To address this challenge, this study introduces the Half-Flip 2D technique, which duplicates the visible half of an occluded face in 2D and merges it into a synthetic frontal image, enabling recovery of key expression features even under vertical occlusion or rotation. The method is evaluated using CNN, DeepFace, FaceLib, and SVM on datasets representing typical online learning scenarios. Experimental results show substantial accuracy improvements in vertical occlusion: 33.79% for CNN, 14.29% for DeepFace, 10.66% for FaceLib, and 6.49% for SVM. For non-frontal poses, improvements reached 16.53% for CNN and 2.21% for SVM. By reducing recognition errors, Half-Flip 2D enhances the stability of emotion-based engagement metrics, thereby supporting more consistent analytics in asynchronous learning environments and enabling adaptive instructional strategies through reliable engagement detection.

INDEX TERMS Engagement, facial emotion recognition, half-flip 2d, occlusion, online learning.

I. INTRODUCTION

Facial Emotions Recognition (FER) has emerged as a significant area of interest within the educational domain, particularly in the context of asynchronous online learning. Asynchronous online learning refers to instructional activities that are not bound by specific time constraints, enabling learners to access and engage with materials at any time and from any location according to their availability [1]. Within this learning mode, FER serves to identify and interpret facial expressions such as happiness, sadness, or confusion in order to evaluate levels of student engagement. In asynchronous environments, FER facilitates the automatic monitoring of learners' emotional states without requiring direct instructor intervention, thereby supporting the provision of more personalized and adaptive feedback.

FER has demonstrated its capability to accurately identify and interpret facial emotions [2]. Nevertheless, several challenges remain, particularly in ensuring that FER systems

can maintain reliable performance when facial features are partially obscured by external factors such as hands, hair, head tilting, or when learners are not directly facing the camera. Such conditions may occur as a result of student disengagement, distractions, or health-related issues that cause discomfort in the neck and head [3]. Under these circumstances, the precision of emotion detection is likely to decline significantly, thereby reducing the overall effectiveness of FER in educational applications.

The problem formulation in this study focuses on addressing the limitations that continue to hinder the effectiveness of Facial Emotions Recognition (FER) in educational contexts. Although FER has proven capable of identifying and interpreting facial emotions, its performance decreases significantly under non-ideal conditions. Specifically, challenges arise when a learner's face is partially occluded by objects (such as hands or hair), tilted, or not directly oriented forward the camera (FIGURE 1).



FIGURE 1. Conditions of hand and head during online learning

These circumstances are common in online learning environments, where students may experience boredom, distractions, or physical discomfort that affects their posture and facial visibility. Consequently, the accuracy of FER in detecting and interpreting emotions is compromised, reducing its reliability as a tool for monitoring engagement and providing adaptive feedback. Therefore, the central problem addressed in this study is how to develop or refine FER systems so that they remain effective in detecting and interpreting facial emotions even when facial visibility is limited or distorted (occluded face).

A variety of strategies have been introduced to mitigate the challenges posed by occlusion in FER. Among these are deep learning (DL) models trained with occlusion-based augmentation to enhance robustness [4], as well as multimodal approaches that integrate facial imagery with complementary data sources such as audio signals [5][6]. While these techniques have demonstrated considerable promise, they frequently rely on computationally intensive processes that can increase latency during FER operations. This underscores the ongoing need for more streamlined yet effective solutions capable of addressing occlusion without incurring substantial computational costs.

An additional component of the FER system pipeline that can be enhanced is the pre-processing stage, particularly through facial reconstruction prior to recognition. Existing studies suggest that reconstruction can be achieved using Generative Adversarial Networks (GANs), which operate by identifying occluding objects, removing them, and subsequently filling the missing regions with plausible facial features derived from previously trained data [7]. While this technique has the potential to improve recognition accuracy, it necessitates training on datasets composed of unobstructed facial images. This requirement limits the practicality of GAN-based reconstruction, especially in resource-constrained environments.

Another potential enhancement in the pre-processing stage of FER systems is facial frontalization, which can be implemented through either 3D reconstruction techniques [8] or 2D transformations [9]. The purpose of frontalization is to reorient a laterally positioned face so that it appears forward-facing. While this method has been shown to mitigate the decline in recognition accuracy associated with non-frontal facial orientations [10], it remains insufficient for addressing occlusion-related challenges. In particular, frontalization cannot effectively handle cases where facial regions are obstructed by external objects, such as hands partially covering the face during online learning sessions.

Based on the findings of Sukumaran et al. (2024), engagement level classification through a multimodal approach that utilizes facial expression data, head orientation, and eye condition has proven effective in achieving accurate detection [11]. This study adopts similar multimodal parameters to those employed in the aforementioned research, but introduces an additional pre-processing stage prior to facial emotion detection. This stage involves the removal of hand occlusion objects and the application of facial frontalization techniques. As a result, the accuracy of FER remains high even when faces are occluded or not in a frontal position, thereby enhancing the overall performance of the engagement detection system. The primary contribution of this study lies in the introduction of a novel pre-processing strategy that integrates frontalization and hand-object removal using 2D image processing prior to emotion classification by the FER model, thereby ensuring robust facial feature extraction and ultimately improving the overall performance of multimodal engagement detection.

II. RELATED WORKS

A. ENGAGEMENT DETECTION IN ONLINE LEARNING

Research on student engagement detection in online learning has increasingly focused on multimodal approaches that utilize diverse data sources to improve system accuracy and reliability. Such approaches extend beyond facial expression analysis, incorporating behavioral, audio, and physiological signals [12]. Evidence suggests that combining facial expression analysis, eye-blink frequency, and head movement enables real-time engagement detection in e-learning sessions. These methods employ deep convolutional neural networks (CNNs) and landmark detection to process webcam data, producing a more accurate engagement index and facilitating immediate feedback for learners [13].

Recent developments include multimodal engagement recognition systems based on webcam imagery that integrate facial expression detection, gaze tracking, head pose estimation, and blink rate analysis through deep learning techniques. Designed specifically for e-learning environments, these systems have demonstrated significant improvements in engagement recognition compared to traditional unimodal models [11]. Overall, the integration of multimodal inputs like visual, audio, and behavioral emerges as a critical strategy for automatically and adaptively detecting and monitoring student engagement in digital learning contexts, thereby establishing a strong foundation for more responsive and personalized online education.

B. FORMS OF OCCLUSION IN FER SYSTEMS

Forms of occlusion in facial emotion recognition (FER) systems include head pose variation [14], illumination variation [15], medical masks [16], hair and glasses [17], [18], or other facial accessories. All these forms of occlusion have the potential to reduce the accuracy of the FER system. Occlusion objects or conditions can either be obtained directly from the dataset (natural) or synthetically created.

Natural occlusion refers to conditions that occur naturally in the real-world environment and are generally uncontrollable by the system, such as head pose variation, lighting variation, and other attributes like hair and glasses. Studies [14], [15], [19], [20], [21], [22], and [8] have shown that changes in head position can significantly block key areas of the face, thus reducing the accuracy of the FER system. Similar findings were also noted in lighting variation [15], which can lead to inconsistency in visual features, as well as personal attributes such as hair and glasses [17], [15], which obscure key areas of the face like the eyes and eyebrows.

Synthetic occlusion refers to conditions that are intentionally created and simulated in experiments to assess the robustness of the FER system under extreme conditions. Common examples include medical masks [16], [23], [24], which cover the mouth and chin areas, masking blocks [25], [26], involving the creation of black block objects to randomly cover areas of the face, and occlusion in specific regions such as the eyes and hair [27], [28]. This type of occlusion is designed to evaluate the impact of covering specific facial areas on FER accuracy, helping to analyze the system's sensitivity to the loss of visual information from the face.

C. APPROACHES IN OCCLUSION FOR FER SYSTEMS

In recent years, research on how to address occlusion issues in FER systems has been dominated by the creation of models that are robust to such occlusions. Models are trained with diverse face data (including occlusion conditions) using public datasets that contain various types of occlusions, such as faces covered by objects, faces turned sideways, or faces under extreme lighting conditions. Some approaches undertaken by researchers include using local attention features [20][29][30], augmenting training data by mixing specific parts of the face [31], or combining several deep learning architectures to build the model, such as CNN+ViT [32] or CNN+LSTM [23].

Another approach researchers have used to overcome occlusion is in the pre-processing stage of the FER system. For instance, Cao et al. took half of the face and combined it in a 3D form [8], followed by muscle enhancement around the nose area to improve emotion recognition accuracy from 3D face data. The results showed improved accuracy for some basic emotions after the muscle enhancement process around the nose. Other 3D processing was also conducted by Ning Sun et al. They performed face frontalization using 3D face landmarks [14], and in general, the accuracy results showed improvements compared to other state-of-the-art methods. However, 3D image processing has limitations in

terms of resource requirements, as it demands higher resources due to more complex computations compared to 2D image processing [9].

Face reconstruction in 2D has also been conducted by some researchers, such as Delphine Poux et al. [28]. They created a synthetic occlusion that covered part of the face with a black object. The method they proposed used optical flow reconstruction, which is a flow that can estimate and fill the occlusion shape, thereby creating a face image with complete features, including the eyes, nose, and mouth. Although the results were not stated to be better than other state-of-the-art methods, the proposed method is competitive in removing occlusions.

Another approach in 2D involves using two flipped (horizontal flip) face images and combining them to appear as a frontal face [22]. This research was conducted by Kai-Yuan Tsai et al. in 2021. The results of the frontalization appeared quite smooth, but the solution they developed was not tested with occlusion types such as hands or other forms, so its reliability in occlusion conditions has not been validated.

D. DATASETS USED IN DEVELOPING FER SYSTEMS WITH OCCLUSION

A wide range of datasets has been employed to study occlusion in facial expression recognition (FER), RAF-DB [33], CK+ [34], AffectNet-7 [35], FERPlus [36], and FER2013 [37] among the most widely used. These datasets provide large collections of facial images annotated with emotion labels, supporting the development and evaluation of robust FER models. CK+, for instance, contains 593 facial expression sequences from 123 individuals, enabling dynamic analysis of expressions from neutral to peak emotions. AffectNet-7, with over 450,000 annotated images, offers one of the most comprehensive resources for emotion recognition, while FER2013 and its enhanced version FERPlus provide benchmark datasets of 48×48 pixel grayscale images, with FERPlus offering refined labels across eight emotion categories.

Beyond general datasets, specialized corpora have been developed to evaluate FER robustness under occlusion and pose variations. Wang et al. (2020) [38] introduced Occlusion-AffectNet, Occlusion-RAF-DB, Occlusion-FERPlus, Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus, specifically designed to simulate real-world challenges such as partial face coverings and non-frontal orientations. These datasets allow researchers to systematically assess how models perform when facial features are obstructed or when faces deviate from frontal alignment.

Occlusion-based datasets introduce artificial or natural obstructions, such as hands, masks, or glasses, to test recognition accuracy when critical facial landmarks are hidden. For example, Occlusion-FERPlus targets occlusions of the eyes, nose, or mouth, while Occlusion-RAF-DB incorporates both synthetic and real-world occlusions. Similarly, Occlusion-AffectNet evaluates model resilience against diverse occlusion scenarios, reflecting practical

conditions encountered in online learning or everyday interactions.

Pose-focused datasets, including Pose-AffectNet, Pose-RAF-DB, and Pose-FERPlus, emphasize variations in head orientation, such as tilted or sideways faces. These resources are crucial for analyzing how pose affects emotion classification accuracy and for developing models capable of generalizing across diverse viewing angles. Collectively, occlusion and pose datasets provide essential benchmarks for advancing FER systems toward greater robustness and applicability in real-world contexts.

III. MATERIALS AND METHOD

A. DATASET

In this work, we utilize the AffectNet and RAF-DB datasets for testing purposes. However, instead of using all the available data from these datasets, we focus on selecting only the relevant data by applying the Py-Feat tool [39] to clean and filter the datasets according to their labels. This ensures that the datasets used for testing contain only well-labeled and accurate facial expressions, thus improving the quality of our evaluation. By carefully curating the data, we aim to minimize any noise or inconsistencies that could affect the results and the reliability of the experiments.

Additionally, to address the varying face orientations in the datasets, we separate frontal faces from side-facing faces using the mediapipe library [40]. This library enables estimation of facial gaze direction, facilitating the categorization of images into frontal and non-frontal groups. This distinction allows for a more thorough evaluation of the proposed method across various real-world conditions, such as side-turned or partially occluded faces. Such categorization is essential for assessing the model's generalization capability in practical contexts, including online learning and video conferencing.

B. PROPOSED METHOD

We propose a pre-processing method called **Half-Face2D**, which is a face frontalization technique that also removes hand occlusions partially covering the face. **FIGURE 2** illustrates the process, starting from an input image (a) to the final output (l). The first step is to perform head roll correction to generate an upright head image (b). The next step is to perform landmarking on image (b), resulting in face landmarks (c) and hand landmarks (d). Then, hand blurring is applied to image (b) based on hand landmarks (d), producing image (e), which will be used as the base image for the final processing stage.

The next step is to mask the face and hand (f), with the face masked in white and the hand masked in red. The resulting masked image will be divided into two parts based on the central points of the face according to the face landmarks (c). The part with the greater number of white pixels will be selected and its middle section will be stretched. Initially, the middle part of the face follows the facial contour, but it will be made upright (with the same X

coordinates). The stretched image will be overlaid with the original face image (c), resulting in image (i). Image (i) is then duplicated and merged to form the complete, unoccluded, and upright face image (l).

Algorithm 1 Half-Flip2D Image Processing

Input:

I_{in} (hand-occluded face or face with side view image)

Step 1: //roll head correction

$I_{rot} \leftarrow \text{RotateForRollCorrection}(I_{in})$

Step 2: //face and hand landmarking

$L_{face}, L_{hand} \leftarrow \text{DetectLandmarks}(I_{rot})$

$I_{src} \leftarrow I_{rot}$

Step 3: //face and hand masking, white for face, red for hand

$M_{face} \leftarrow \text{CreateFaceMask}(I_{rot}, L_{face}, \text{WHITE})$

if $L_{hand} = \text{TRUE}$ then

$M_{hand} \leftarrow \text{CreateHandMask}(I_{rot}, L_{hand}, \text{RED})$

$I_{src} \leftarrow \text{ApplyHandBlurring}(I_{rot}, L_{hand})$

$M_{full} \leftarrow \text{CombineMasks}(M_{face}, M_{hand})$

else

$M_{full} \leftarrow M_{face}$

end if

Step 4: //divide face to two region

$M_{left}, M_{right} \leftarrow \text{DivideByCenter}(M_{full}, L_{face})$

$CM_{left} \leftarrow \text{CountWhitePixels}(M_{left})$

$CM_{right} \leftarrow \text{CountWhitePixels}(M_{right})$

Step 5: //compare two region of the face

if $CM_{left} > CM_{right}$ then

$M_{sel} \leftarrow M_{left}$

Else

$M_{sel} \leftarrow M_{right}$

end if

Step 6: //stretch the center poin of the selected region face

$M_{stretch} \leftarrow \text{StretchToStraightLine}(M_{sel})$

Step 7: //replace masked face with the original face source

$I_{sel} \leftarrow \text{ReplaceRegion}(I_{src}, M_{stretch})$

Step 8: //duplicate and flip the image

$I_{flip} \leftarrow \text{DuplicateFlipHorizontal}(I_{sel})$

Step 9: //combine output of step 7 and step 8, then resize

$I_{full} \leftarrow \text{Combine}(I_{sel}, I_{flip})$

$I_{out} \leftarrow \text{ResizeImage}(I_{full}, 224, 224)$

Output: I_{out} (non-occluded and frontal face)

The flow of the proposed method is further detailed in Algorithm 1. The "DetectLandmark" function in Algorithm 1 serves as a substitute for traditional face detection techniques such as MTCNN, RetinaFace, or Haarcascade models. The underlying logic is that if facial landmarks are not detected, the image will be classified as not containing a face. This function plays a critical role in ensuring the accuracy and efficiency of the proposed method. For a clearer understanding of the process, the visualization of the proposed method can be observed in **FIGURE 2**.

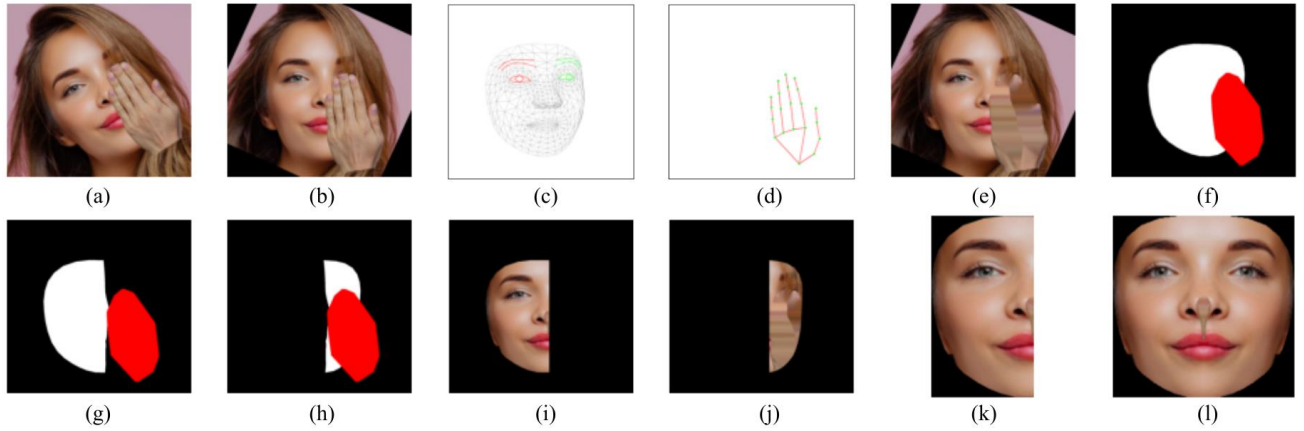


FIGURE 2. Steps for processing of proposed method, start from input image (a) and finish with the output (l)

IV. METHODOLOGY

A. DATA SELECTION

The first step in our methodology involves face selection from the datasets. We began by combining the AffectNet and RAF-DB datasets to create a comprehensive data pool for testing. However, given the presence of inconsistencies and mislabeled data in the AffectNet dataset, it was necessary to apply a cleaning procedure. To address this, we utilized the Py-Feat tool [39][39], which facilitated the filtering and selection process, ensuring that only the correctly labeled and relevant facial data were retained for further analysis. This cleaning process is crucial, as it removes any erroneous or misclassified samples, which could otherwise compromise the integrity of the model's performance and the validity of our results. The outcome of this cleaning process is illustrated in FIGURE 3, which demonstrates the refinement of the dataset to ensure accurate and reliable testing.



FIGURE 3. Inaccurate facial annotations in the AffectNet dataset

After that, we separated the dataset into frontal face data and non-frontal face data. The separation was done by calculating the yaw angle of the head using the mediapipe library. If the yaw angle is less than 30 degrees, it is considered a frontal face, otherwise it is considered a non-frontal face. FIGURE 4 shows the results of the selection process.



FIGURE 4. (a) Frontal face (b) Non-frontal face

B. HAND SYNTHESIS

To evaluate the effectiveness of the proposed method in removing hand occlusions and performing frontalization, this study requires a dataset with vertical hand occlusion covering half of the face. We conducted augmentation by randomly adding synthetic hands to the frontal face images as shown in FIGURE 5.



FIGURE 5. (a) Raw image (b) Hand synthetic image (c) Output image

This study involves two types of data for testing, face with partial hand occlusion and non-frontal face, as shown in FIGURE 6. The dataset includes 1,400 images of faces with partial hand occlusion and 1,043 images of side-facing faces, with the distribution of each emotion detailed in TABLE I.

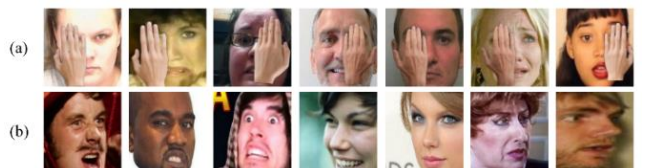


FIGURE 6. Testing data (a) Faces with hand occlusion (b) Faces with side view

TABLE I
TEST DATA DISTRIBUTION

Fig. 6	An	Di	Fe	Ha	Ne	Sa	Su
(a)	200	200	200	200	200	200	200
(b)	72	138	98	200	200	157	178

C. PROPOSED METHOD IMPLEMENTATION

The next step is to implement the proposed method on the test data (FIGURE 6) as described in Algorithm 1 and illustrated in FIGURE 5. The testing was conducted on the entire prepared test dataset in RGB image format with various background conditions. The output remains in RGB format, but the background is removed and replaced with a black background (FIGURE 7), ensuring that the resulting image focuses solely on the face shape before classification with the FER model.

D. FER MODEL

We also developed a FER model with a Convolutional Neural Network (CNN) architecture using only frontal faces from the AffectNet dataset, with 350 images per emotion class (FIGURE 7). The goal was to test the impact of facial occlusion, specifically faces partially covered by hands and faces in non-frontal positions.

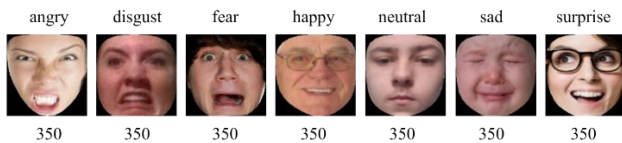


FIGURE 7. Training data

The dataset was split into 80% for training and 20% for validation. The model training was conducted using the early stopping technique, which aims to automatically stop the training process if the validation accuracy does not show improvement or even decreases. This method helps prevent overfitting by ensuring that the model does not train too long on the training data, which could lead to performance degradation on the validation data. The training process utilized the Adam optimizer with an initial learning rate of 0.001, a batch size of 32, and a maximum of 50 epochs, adjusted to ensure stable model convergence without sacrificing generalization.

E. EVALUATION

The evaluation was conducted by measuring the accuracy of facial emotion recognition by calculating the difference in accuracy between the data before and after the proposed method was applied. This approach was used to assess the impact of the proposed method, which may either improve or decrease the accuracy.

We used four FER models to test the proposed method. First, testing was conducted with the custom CNN model, which was trained exclusively on frontal face data, followed by testing with DeepFace [41], FaceLib [42], and SVM. The testing was conducted separately between the data with hand occlusion and the data with non-frontal faces. Specifically, for non-frontal faces, we also compared the results from our proposed method with the results from the 2D frontalization method performed by Kai-Yuan Tsai et al. (2021) [22]. The testing on the data with hand occlusion and non-frontal face was conducted using the combination of the RAF-DB and

AffectNet datasets. WAdditionally, we will conduct testing on data with horizontal hand occlusion (covering the lower part/mouth), a common occurrence in real-world scenarios, as shown in FIGURE 8.



FIGURE 8. Horizontal hand-occlusion face

F. ENGAGEMENT CALCULATION

We will integrate the results of the proposed method into the implementation of FER within an online learning system using emotion data, head orientation, and eye gaze [11]. The weighting (TABLE II – IV) is applied to determine the students' engagement status using an engagement index.

TABLE II
WEIGHT OF EMOTIONS

Emotions	Ang	Dis	Fea	Hap	Neu	Sad	Sur
Weight	0.1	0.9	0.5	1.1	1.4	0.3	0.7

TABLE III
WEIGHT OF GAZE DIRECTION

Look Center	Not Look Center
1	0

TABLE IV
WEIGHT OF EYE STATUS

Eye Open	Eye Close
2.5	0

The Egement Index (EI) will be calculated for each facial frame using Equation 1.

$$EI = g(i) + e(i) + f(i) \quad (1)$$

where EI is the Engagement Index value, $g(i)$ is the weight of the gaze direction, $e(i)$ is the weight of the eye status, and $f(i)$ is the weight of the facial emotion obtained from the FER model classification. Each frame will have its engagement status determined using the rules in TABLE V. The overall conclusion of a student's engagement will be determined by the most dominant engagement status. An overview of this entire work can be seen in FIGURE 9.

TABLE V
ENGAGEMENT STATUS CLASSIFICATION BASED ON EI VALUE

Engagement status	EI
Highly Engaged	$EI \geq 4.5$
Confused	$4 \leq EI < 4.5$
Boredom	$2.5 \leq EI < 4$
Sleepy	$0 < EI < 2.5$
Very Not Engaged	$EI = 0$

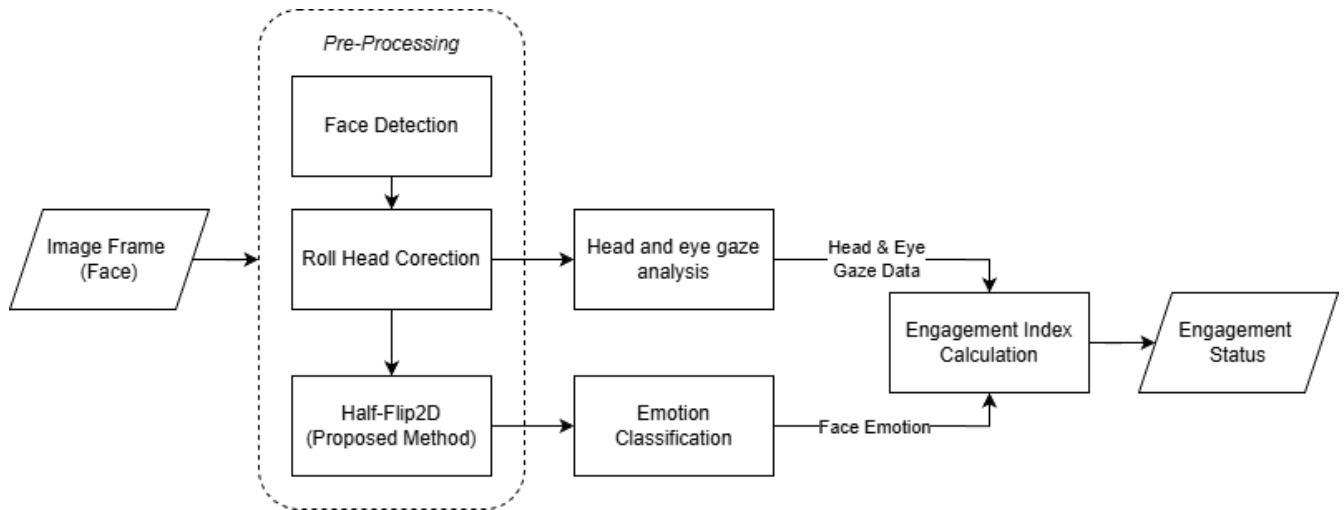


FIGURE 9. The workflow of the engagement assessment system in online learning

V. RESULT

A. CNN MODEL RESULT

Based on the experimental results of creating the FER model with a CNN architecture, the training was stopped after 26 iterations, with a training accuracy of 0.88 and a validation accuracy of 0.65, as shown in FIGURE 10. This result indicates that while the model achieved good accuracy on the training data, its ability to generalize to validation data still needs improvement.

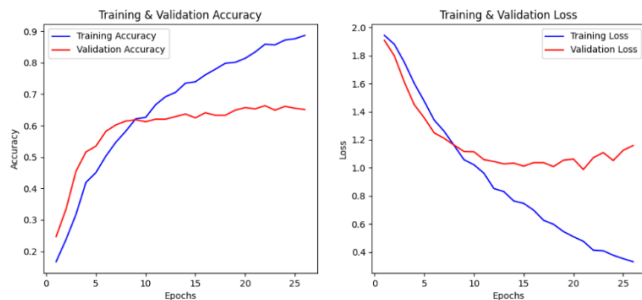


FIGURE 10. Train and validation accuracy result

This condition was deliberately created to test the model's weaknesses in validation with face data partially occluded by a vertical hand and non-frontal face conditions, thereby providing an alternative approach to address the limitations of an emotion recognition model in online learning through pre-processing techniques as proposed in our method.

Based on the results of the confusion matrix (FIGURE 11) obtained from the emotion expression recognition model trained using only frontal face data without augmentation, it can be concluded that the model shows fairly good accuracy on some dominant emotion classes, such as happy and angry. However, there is still overlap in classification among classes with similar visual features, such as fear, surprise, and neutral. This aligns with research conducted by Szczakowska et al. (2023) [43], which emphasizes that without data augmentation and diversification, models tend to be biased toward the majority class and struggle to distinguish emotions with similar

features, thus leading to suboptimal classification performance for minority classes. Therefore, the application of data augmentation strategies is recommended to improve generalization ability and reduce classification errors between classes, as highlighted in recent studies related to facial emotion recognition based on deep learning.

As an alternative to data augmentation, which can increase computational load and extend training time [44], this study adopts a pre-processing approach before the emotion recognition process by the model. By performing standardization, normalization, and other transformations on facial images prior to the classification stage, the pre-processing technique has been empirically shown to improve both the accuracy and efficiency of the emotion recognition system, as well as help reduce classification errors between classes [45].

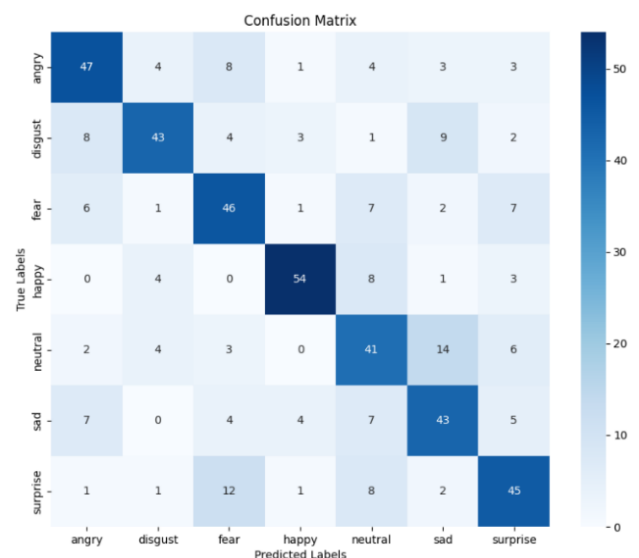


FIGURE 11. Confusion matrix of CNN model

B. PROPOSED METHOD RESULT

This section presents the results of the face pre-processing experiments conducted on images featuring hand occlusion

and non-frontal poses, as outlined in the proposed method. The pre-processing pipeline aims to address common challenges encountered in real-world scenarios, where faces may be partially covered or not oriented directly towards the camera. By simulating these challenging conditions, the evaluation provides insight into the effectiveness of the developed approach.

FIGURE 12 demonstrates the outcome of the pre-processing stage for an image exhibiting both hand occlusion and a non-frontal pose. The input image is systematically processed to mitigate occlusion effects and to align the facial region prior to classification by the FER model. Such pre-processing is crucial for enhancing the accuracy of emotion recognition, as it improves the model's ability to extract meaningful features despite the presence of occlusions or pose variations.

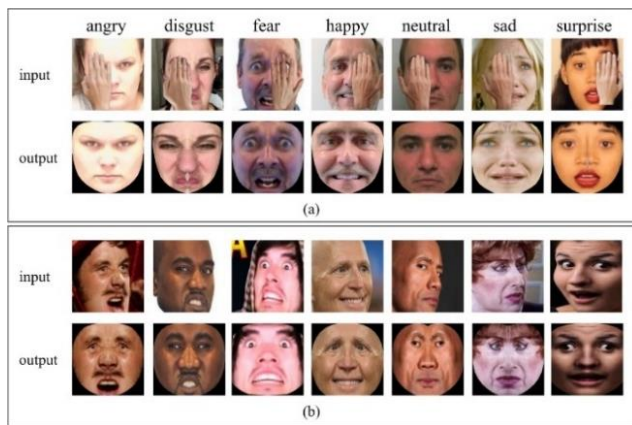


FIGURE 12. Proposed method result (a) Test result with hand-occluded face images (b) Test result with side view face images

The probabilities assigned to both the input image (“before”) and output image (“after”) were evaluated using the CNN model trained exclusively with frontal face data. As shown in FIGURE 13, a face labeled as “angry” in the ground truth initially exhibited a dominant model prediction of “fear” prior to pre-processing. However, after applying the proposed pre-processing method, there was a significant increase in the probability for “angry,” successfully shifting the dominant emotion classification from “fear” to “angry” and thereby matching the ground truth label. The detailed probability scores for each emotion are provided in TABLE VI to substantiate these findings.

These results confirm the efficacy of the proposed approach in enhancing emotion recognition accuracy, particularly for faces affected by occlusions or non-frontal poses. By refining the input image through pre-processing, the model demonstrates an improved ability to correctly classify emotions even under challenging conditions. Moreover, this pre-processing strategy contributes to the interpretability and reliability of FER model outputs by aligning input images with the model's training distribution, reducing the likelihood of misclassification due to pose variations or occlusions. Consequently, this methodological

improvement supports more robust FER performance not only in controlled experiments but also in practical applications, where variations in facial orientation and partial occlusions are frequently encountered.

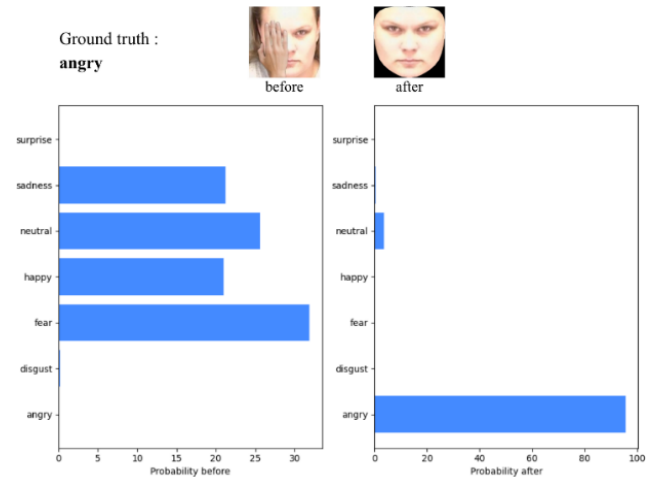


FIGURE 13. Probability of before and after proposed method implementation

FIGURE 13 illustrates the probability values for each emotion class before and after applying the proposed pre-processing method. The emotion class with the highest probability value (dominant emotion) corresponds to the output of the facial emotion recognition (FER) model, demonstrating that even a relatively small difference in probability scores can significantly impact the final classification outcome. This finding underscores the importance of precise probability adjustment in achieving reliable emotion recognition results.

The test results using the face image presented in FIGURE 13 reveal a substantial increase in the probability of the “Angry” emotion, which initially registered a value of 0.028 and subsequently rose to 95.612 after implementation of the proposed method. This dramatic shift indicates a marked improvement in emotion classification, as the dominant model prediction switched from “Fear” (Fe) to “Angry” (An), thereby aligning more closely with the ground truth label. The magnitude of the probability change (an increase of 95.584) further confirms the effectiveness of the pre-processing approach in enhancing classification accuracy for challenging cases.

To provide a more comprehensive understanding, the following section will discuss the broader impact of the proposed method on four leading FER approaches: CNN, DeepFace, FaceLib, and SVM. Performance comparisons will be conducted based on the test data depicted in FIGURE 6, with detailed quantities summarized in TABLE I, offering a robust evaluation of method generalizability and practical utility across multiple FER systems.

TABLE VI
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH CNN

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	34.00	74.00	40.00	Increased
Disgust	100	2.00	45.50	43.50	Increased
Fear	100	37.50	56.50	19.00	Increased
Happy	100	33.50	71.50	38.00	Increased
Neutral	100	2.00	61.50	59.50	Increased
Sad	100	31.00	51.50	20.50	Increased
Surprise	100	33.00	49.00	16.00	Increased
Average				33.79	Increased

TABLE VII
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH DEEPPFACE

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	36.50	38.00	1.50	Increased
Disgust	100	2.00	9.00	7.00	Increased
Fear	100	26.50	26.50	0.00	No Change
Happy	100	16.00	70.00	54.00	Increased
Neutral	100	19.00	64.00	45.00	Increased
Sad	100	29.50	35.50	6.00	Increased
Surprise	100	35.50	22.00	-13.50	Decreased
Average				14.29	Increased

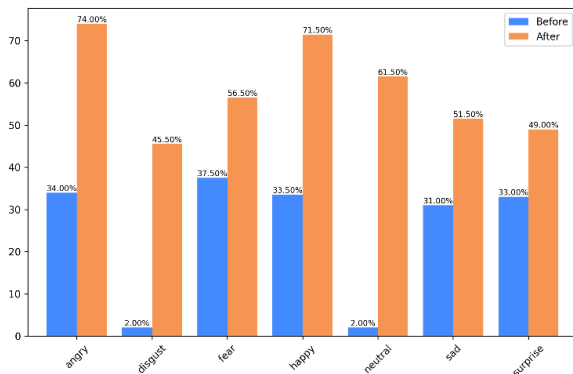


FIGURE 14. The result of implementing the proposed method on the hand-occluded face using the CNN model

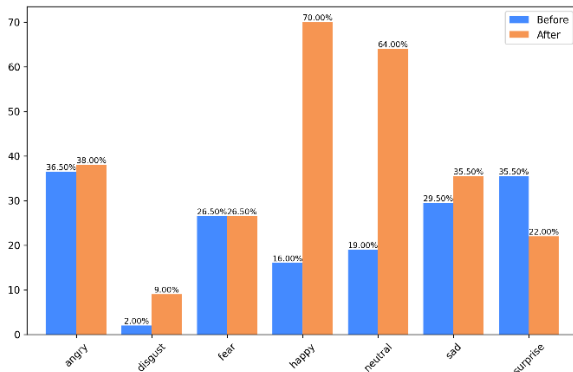


FIGURE 15. The result of implementing the proposed method on the hand-occluded face using the Deepface model

TABLE VIII
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH FACELIB

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	76.00	40.79	50.00	9.21	Increased
Disgust	88.00	3.41	3.98	0.57	Increased
Fear	99.50	48.24	65.83	17.59	Increased
Happy	99.50	45.23	77.39	32.16	Increased
Neutral	99.50	22.11	41.21	19.10	Increased
Sad	99.50	55.78	66.83	11.05	Increased
Surprise	99.50	51.26	36.18	-15.08	Decreased
Average				10.66	Increased

TABLE IX
ACCURACY RESULT OF HAND-OCCLUDED FACE WITH SVM

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	80.50	66.00	14.50	Increased
Disgust	100	0.50	0.50	0.00	No Change
Fear	100	70.00	54.00	-16.54	Decreased
Happy	100	36.00	81.50	45.50	Increased
Neutral	100	8.00	8.50	0.50	Increased
Sad	100	12.00	45.00	33.00	Increased
Surprise	100	3.00	0.50	-2.50	Decreased
Average				6.49	Increased

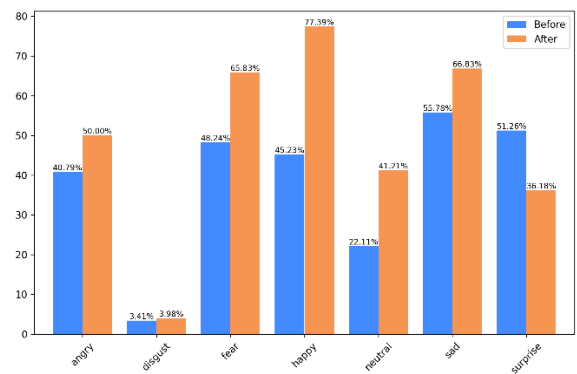


FIGURE 16. The result of implementing the proposed method on the hand-occluded face using the Facelib model

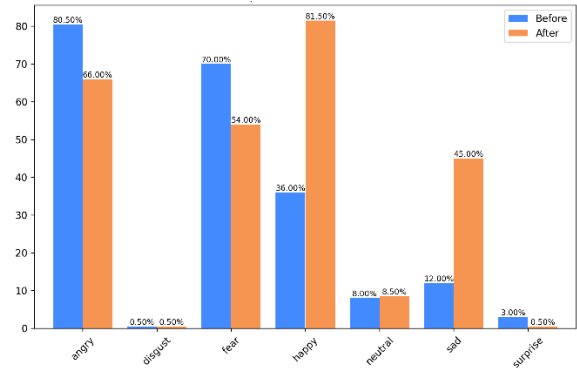


FIGURE 17. The result of implementing the proposed method on the hand-occluded face using the SVM model

TABLE X
ACCURACY RESULT OF NON-FRONTAL FACE WITH CNN

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	76.39	50.00	-26.39	Decreased
Disgust	100	2.90	14.49	11.59	Increased
Fear	100	17.35	45.92	28.57	Increased
Happy	100	18.00	32.50	14.50	Increased
Neutral	100	4.50	33.00	28.50	Increased
Sad	100	21.66	73.89	52.23	Increased
Surprise	100	27.53	34.27	6.74	Increased
Average			16.53	Increased	

TABLE XI
ACCURACY RESULT OF NON-FRONTAL FACE WITH DEEPPFACE

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	34.72	36.11	1.39	Increased
Disgust	100	2.90	0.72	-2.18	Decreased
Fear	100	27.55	19.39	-8.16	Decreased
Happy	100	70.00	61.5	-8.50	Decreased
Neutral	100	42.50	56.50	14.00	Increased
Sad	100	29.50	45.22	-3.19	Decreased
Surprise	100	23.03	22.47	-0.56	Decreased
Average			-1.03	Decreased	

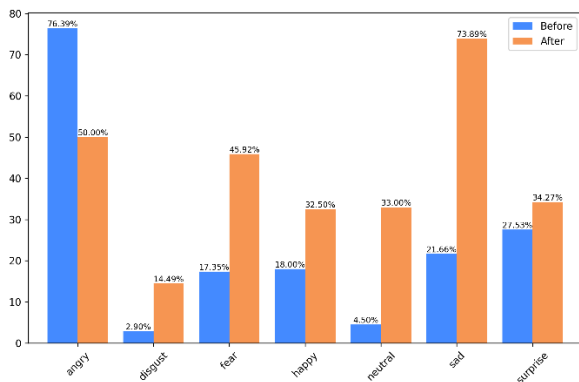


FIGURE 18. The result of implementing the proposed method on the non-frontal face using the CNN model

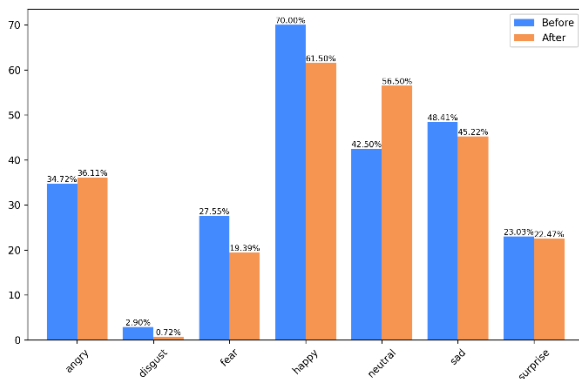


FIGURE 19. The result of implementing the proposed method on the non-frontal face using the Deepface model

TABLE XII
ACCURACY RESULT OF NON-FRONTAL FACE WITH FACELIB

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	70.83	51.11	-9.72	Decreased
Disgust	99.27	1.46	0.73	-0.73	Decreased
Fear	100	22.45	48.98	26.53	Increased
Happy	99.50	74.87	53.77	-21.10	Decreased
Neutral	99.50	46.73	31.66	-15.10	Decreased
Sad	98.08	55.19	67.63	12.44	Increased
Surprise	97.75	54.02	25.86	-28.20	Decreased
Average			-5.12	Decreased	

TABLE XIII
ACCURACY RESULT OF NON-FRONTAL FACE WITH SVM

Emotion	Face Detected (%)	Before (%)	After (%)	Difference (%)	Conclusion
Angry	100	72.22	69.44	-2.78	Decreased
Disgust	100	0.00	0.00	0.00	No change
Fear	100	68.37	63.27	-5.10	Decreased
Happy	100	56.50	68.00	11.50	Increased
Neutral	100	14.50	5.00	-9.50	Decreased
Sad	100	14.65	38.22	23.57	Decreased
Surprise	100	3.93	1.69	-2.24	Decreased
Average			2.21	Increased	

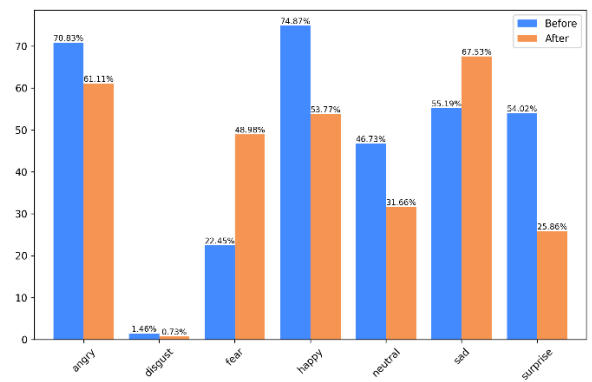


FIGURE 20. The result of implementing the proposed method on the non-frontal face using the Facelib model

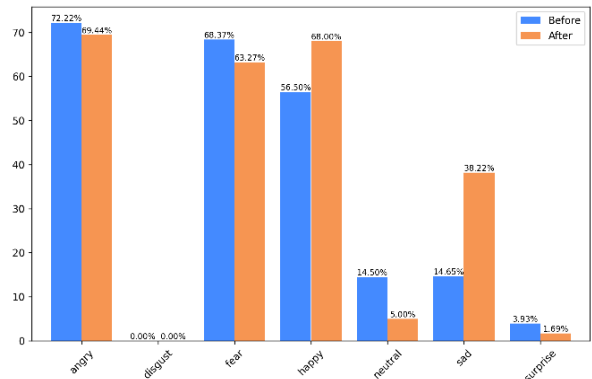


FIGURE 21. The result of implementing the proposed method on the non-frontal face using the SVM model

In addition to testing on data with vertical hand occlusion covering the face, we also evaluate the proposed method on data with horizontal hand occlusion covering the mouth area, as shown in [FIGURE 8](#). This scenario, commonly encountered in real-world situations, is tested using the carefully filtered RAF-DB dataset. Importantly, this dataset was not used in the training phase of the CNN model, which was trained solely on frontal face data. This approach ensures the model is tested on unseen data, providing an accurate assessment of its performance in real-world conditions with varying occlusions.

TABLE XIV
SELECTED RAFDB DATA DISTRIBUTION

An	Di	Fe	Ha	Ne	Sa	Su
106	62	73	200	200	200	200



FIGURE 22. The result of applying the proposed method to a face with hand occlusion in the mouth area

TABLE XV
DETECT FACE PERFORMANCE OF THE MODELS WITH HAND OCCLUSION IN THE MOUTH AREA USING SELECTED RAFDB DATA

Emotion	CNN (%)	DeepFace (%)	FaceLib (%)	SVM (%)
Angry	100	100	96.22	100
Disgust	100	100	100	100
Fear	100	100	93.15	100
Happy	100	100	99.50	100
Neutral	100	100	99.50	100
Sad	100	100	99.50	100
Surprise	100	100	99.50	100

TABLE XVII
COMPARISON RESULT

Emotion	CNN (%)		DeepFace (%)		FaceLib (%)		SVM (%)	
	A	B	A	B	A	B	A	B
Angry	-34.62	-26.39	-7.70	1.39	-14.29	-9.72	3.85	-2.78
Disgust	12.93	11.59	-3.40	-2.18	-2.04	-0.73	-0.68	0.00
Fear	17.82	28.57	-10.89	-8.16	16.83	26.53	-30.7	-5.10
Happy	29.00	14.50	-6.00	-8.50	4.53	-21.10	22.00	11.5
Neutral	28.00	28.50	17.50	14.00	-5.02	-15.07	1.00	-9.50
Sad	27.10	52.23	-15.49	-3.19	6.50	12.44	20.64	23.57
Surprise	25.34	6.74	-6.00	-0.56	-20.41	-28.16	-2.00	-2.24
Average	15.08	16.53	-4.57	-1.03	-1.99	-5.12	2.02	2.21

Positive = increase;
Negative = decrease;
A = Other method [22];
B = Our Method.

TABLE XVI

AVERAGE ACCURACY CHANGES BEFORE AND AFTER APPLYING THE METHOD ON FACES WITH HAND OCCLUSION IN THE MOUTH AREA

Emotion	CNN (%)	DeepFace (%)	FaceLib (%)	SVM (%)
Angry	-2.83	4.72	-0.98	-9.44
Disgust	-1.61	-3.23	0.00	0.00
Fear	-12.33	-5.48	14.71	0.00
Happy	-4.00	-6.50	4.02	-2.00
Neutral	32.00	-9.50	-41.70	-10.5
Sad	-21.5	-11.00	12.57	16.5
Surprise	12.00	10.5	-9.04	-0.5
Average	0.25	-2.93	-2.92	-0.85

Positive = increase;
Negative = decrease

C. COMPARISON WITH STATE-OF-THE-ART METHOD

The evaluation will also include a comparison of our method with the frontalization technique performed by K.Y. Tsai et al. (2021) [22]. They also conducted frontalization using only 2D image processing. This comparison aims to assess the relative effectiveness of our proposed method in handling facial occlusions and non-frontal faces compared to the existing frontalization approach. By benchmarking both methods, we can evaluate the strengths and weaknesses of each in terms of accuracy and robustness in real-world applications.



FIGURE 23. Comparison result, (A) input image, (B) our method, (C) other method [22]

D. ENGAGEMENT DETECTION RESULT

In this section, the results of integrating the proposed method with the FER system in online learning specifically in asynchronous learning will be presented, where student engagement is evaluated while they watch the instructional video provided by the teacher. FIGURE 24 shows a condition in which a student is watching the material presented by the instructor.

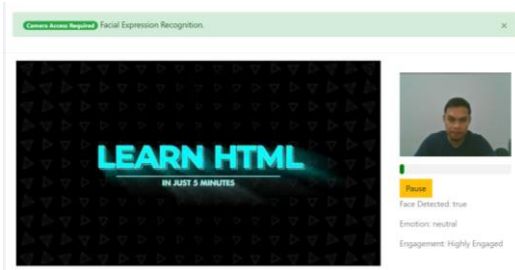


FIGURE 24. A page in asynchronous learning that contains the video material frame and the FER system frame

The system will produce an insight containing the student's engagement data, emotion percentages, and the engagement conclusion based on the dominant engagement status across the duration of the data, which corresponds to the length of the instructional video provided by the teacher. A visualization of the results can be seen in FIGURE 26.

We tested three students to evaluate their engagement levels. The first student was scripted to be predominantly bored, the second to be predominantly confused, and the third to be predominantly highly engaged (FIGURE 25).

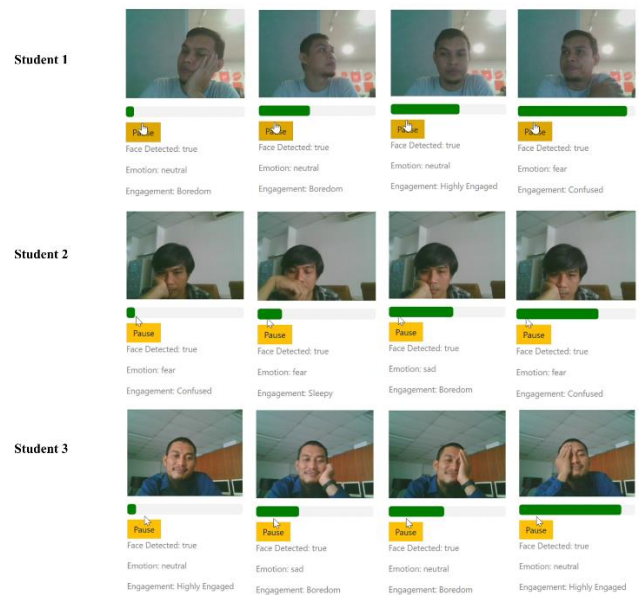


FIGURE 25. Students condition in asynchronous learning scenario

TABLE XVIII
STUDENT ENGAGEMENT COMPOSITION

Student	A	B	C	D	E
Student 1	20.00	11.11	53.33	8.89	6.67
Student 2	13.33	51.11	22.22	11.11	2.22
Student 3	51.11	2.22	35.56	6.67	4.44

A = Highly engaged; B = Confused; C = Boredom; D = Sleepy; E = Very not engaged

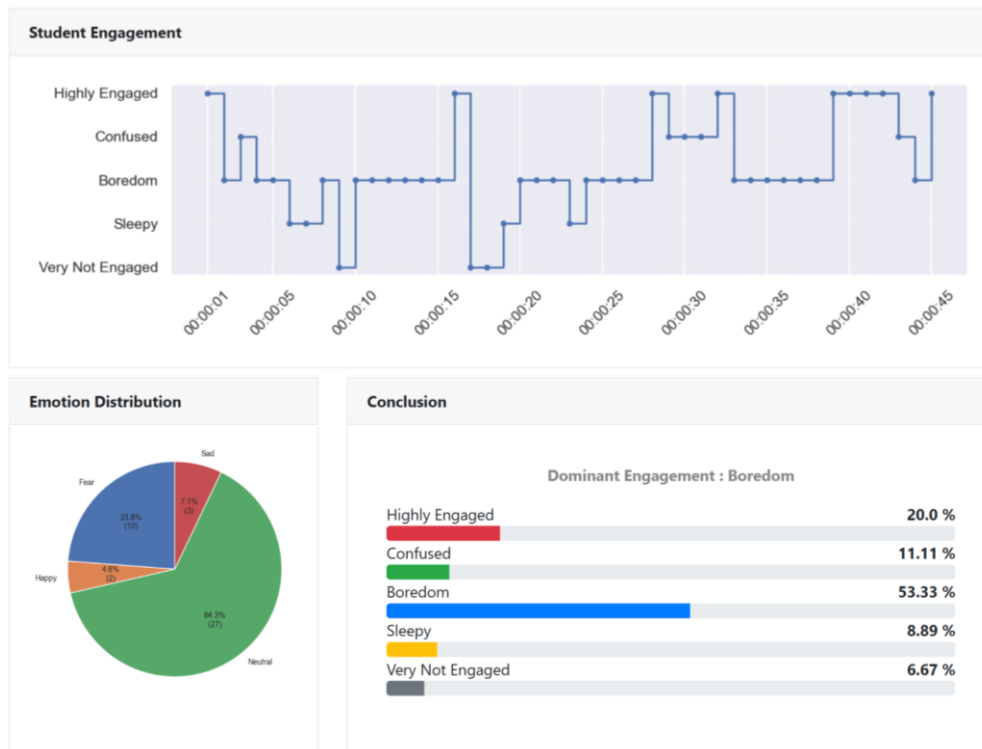


FIGURE 26. A page that contains the student's engagement insight results

VI. DISCUSSION

A. FACE DETECTOR

The first stage in a Facial Expression Recognition (FER) system is face detection. If this stage fails, the entire subsequent pipeline will stop because the system cannot extract expression features from images that are not identified as faces. According to the research conducted using four state-of-the-art methods, CNN, DeepFace, Facelib, and SVM, only the Facelib method failed in face detection, while the other three methods successfully recognized faces 100 percent of the time. Facelib uses the Single Shot Multibox Detector (SSD) algorithm for face detection.

SSD is a one-stage detector architecture that combines CNN-based feature extraction with a multi-scale feature map mechanism to detect objects of various sizes simultaneously. Its main advantage lies in its high inference speed, as both classification and bounding box regression are performed in one step. However, several studies have shown that this efficiency comes at the cost of precision, especially under conditions of occlusion or pose variation [46]. SSD performs optimally on frontal face images and those without background interference, but its performance drastically decreases when parts of the face are occluded, for example, by hands or masks, or when the face is in a non-frontal position.

The results presented in TABLE VIII and TABLE XII support this theory, where Facelib with the SSD detector achieved only 76 % performance on hand-occluded faces in the "angry" class and 88 % in the "disgust" class, while other classes, including fear, happy, neutral, sad, and surprise, showed higher results, around 99.5 %. For non-frontal face images, the detection performance for each class was 100 percent, 99.27 %, 100 %, 99.50 %, 99.50 %, 98.08 percent, and 97.75 %. The performance drop in detecting occluded faces aligns with the findings of Jang et al. (2019), who demonstrated that the Face-SSD algorithm variant can only maintain accuracy in the range of 80 to 95 percent, depending on the degree of occlusion [47].

Thus, although other studies have suggested that SSD offers superior real-time performance [48], the results in this study indicate that, without improvements in the architecture and pre-alignment stages, SSD's performance remains inadequate for FER applications in real-world environments that are characterized by occlusions and pose variations.

B. PARTIAL OCCLUSION BY HAND

The presence of hands partially covering the face is one of the most common forms of partial occlusion in real-world scenarios, such as when someone reacts spontaneously to a specific emotion or experiences boredom during online lessons, where hands are often used to support the head or face. This type of occlusion significantly reduces the performance of Facial Expression Recognition (FER) systems as it obstructs key facial areas that are crucial in determining expressions, such as the eyes, nose, and particularly the mouth. When these areas are missing from the input image, the features extracted during the process

become incomplete, making it difficult for the model to map the patterns to the correct emotion.

Experimental results clearly demonstrate this effect. On the test data before pre-processing with the proposed method (the "before" data), the average FER accuracy was only 24.71%, 23.57%, 38.12%, and 30.08% for the CNN, DeepFace, Facelib, and SVM models, respectively. These results indicate that nearly two-thirds of the classifications failed to recognize emotions correctly when the face was covered by a hand. This issue was particularly evident in deep learning-based models such as CNN and DeepFace, where despite the model's ability to learn complex features, it remained highly dependent on the completeness of the facial information.

After applying the pre-processing with the proposed method to remove the hand occlusion from the facial images, there was a significant increase in accuracy. The accuracy rose to 58.5%, 37.9%, 48.8%, and 36.6% for the four models. The average increase of over 20% demonstrates the effectiveness of the pre-processing stage in recovering important information that was previously lost. This process helps the model generate more stable feature representations, even when part of the face remains occluded by the hand.

Neutral and happy emotions were the most positively impacted by the proposed method under conditions of vertical hand occlusion. The sharp increase in accuracy for neutral emotions (up to +59.5% with CNN and +45% with DeepFace) indicates that the representation of neutral expressions greatly benefits from the proposed pre-processing in the areas obstructed by the hand. This might be due to the less prominent dynamic features of neutral expressions, such as the mouth, making it easier for the model to recognize the main structure of the face, even when partially covered. A similar sharp increase was observed for happy emotions, with DeepFace (+54%), FaceLib (+32.16%), and SVM (+45.5%). This suggests that visual indicators of happiness, such as cheek movement and contraction of the orbicularis oculi around the eyes, can still be captured by the model, even when parts of the face are covered by a hand.

On the other hand, emotions such as surprise and fear were most negatively impacted, with a notable decrease in accuracy, especially for the DeepFace and FaceLib models. This decrease indicates that the expression features for these two emotions, such as eye widening or sudden changes in the upper face, are highly susceptible to disruption due to hand occlusion on the upper portion of the face. With a hand covering parts of the eyes or forehead, the characteristic patterns of surprise and fear expressions become difficult for the model to recognize. This finding underscores the importance of the upper face in detecting highly dynamic emotions and highlights the need for developing algorithms that are more sensitive to the presence of occlusions in key expressive areas, such as the eyes and eyebrows.

From a system interpretation perspective, this improvement also indicates that a robust FER approach to occlusion requires explicit handling at the face detection or

normalization stages. In other words, the model's architecture is not the only factor determining the final performance, but also the quality and integrity of the input received. In real-world applications such as smart surveillance, driver monitoring, and emotion-based interaction, the issue of hands covering the face must be considered from the design stage of the pipeline so that the system can remain reliable, even when encountering faces that are not fully exposed.

There is a limitation in the proposed method, where the hand detection model from MediaPipe occasionally fails to detect the presence of a hand on the face, resulting in a failure in subsequent processing stages, leading to outputs that do not meet expectations (FIGURE 27). The number of hand detection failures for each emotion class was as follows: angry 9, disgust 6, fear 13, happy 17, neutral 11, and surprise 17, from a total of 200 images per emotion class. Therefore, it can be concluded that the hand detection failure rate is below 0.1%.

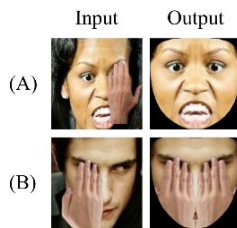


FIGURE 27. (A) Success condition (B) Fail condition

The failure of the hand detection model from MediaPipe in detecting the presence of hands on the face is generally caused by a combination of technical factors and complex visual conditions. One of the main reasons is the overlap between the hand and the face, where similar skin tones and the hand being too close to the face make it difficult for the system to distinguish the hand area from the face's contour [49]. Image quality and lighting conditions also play a crucial role, as uneven lighting, shadows on the face, or high reflections can lower the confidence score below the threshold of 0.7 set by the MediaPipe model. Failures also frequently occur when the hand size in the image is too large or too small, due to the camera distance not matching the ideal detection ratio.

C. NON-FRONTAL FACE (POSE-INVARIANT)

Another form of occlusion tested in this study is the condition of non-frontal faces or pose-invariant faces. While the pre-processing with the proposed method successfully increased the accuracy of the FER model on faces partially

occluded by hands (vertically), this pre-processing also led to a decrease in accuracy for FER models on non-frontal face conditions. The experimental results TABLE VIII – X) show that for the "before" data, the accuracy for the CNN, DeepFace, FaceLib, and SVM models were 24.05%, 35.59%, 46.51%, and 32.88%, respectively. After applying the pre-processing with the proposed method, the accuracies increased to 40.6%, 34.6%, 41.4%, and 35.1%. The positive impact was observed only in the CNN and SVM models, while the accuracy of the DeepFace and FaceLib models decreased. The increase of over 15% in the CNN model is likely due to the fact that the model was trained using only frontal face data, so the test data, which was frontal after pre-processing, was easily recognized by the model.

For DeepFace, a decrease in accuracy of -1.03% occurred, which may be due to the fact that DeepFace (FIGURE 28) already applies a frontalization pre-processing step in its architecture. As a result, when tested with the proposed method, there was a double frontalization process—first by the proposed method, then again by DeepFace. This double processing led to redundant spatial transformations, where the face, already frontalized, was processed again at the internal stage of DeepFace, causing minor deformations in the facial structure, which actually reduced classification accuracy. However, this decrease in accuracy was minimal and did not significantly affect the overall model performance. The same issue occurred with FaceLib, where FaceLib already incorporates face alignment in its pipeline, as indicated in the FaceLib Python code documentation on its official GitHub link.

The frontalization process applied in this study aims to normalize the face orientation to align with the camera's view. However, this process has the potential to cause side effects, such as texture smoothing on specific areas of the face [50]. In conditions where one side of the face is covered by a hand or there is uneven lighting distribution, this replication process can create artificial textures in those areas. On the other hand, these artificial textures may alter the authenticity of the local feature patterns around the reconstructed area, thereby affecting the image representation used by the emotion recognition system. In this context, FaceLib, which relies on spatial and textural information to extract expression features, may misinterpret these replicated patterns as characteristics of an emotion different from the actual condition. As a result, the emotion classification process may experience a degradation in accuracy due to the mismatch in texture structure resulting from the frontalization process.

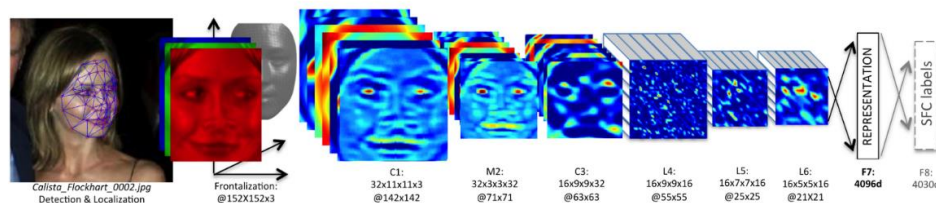


FIGURE 28. DeepFace Architecture [41]

In line with the comparison results between the proposed method and the frontalization method by K.Y. Tsai et al. (2021) [22] presented in TABLE XIV, the findings indicate that both the proposed method and the comparison method lead to similar conclusions. Specifically, there is an increase in accuracy for CNN and SVM models, while a decrease in accuracy is observed in the DeepFace and FaceLib models. As previously explained, DeepFace and FaceLib already incorporate face alignment within their pipeline, meaning that applying pre-processing before using these models can potentially lower their accuracy.

Overall, these results demonstrate that the impact of pre-processing on model performance largely depends on the design and characteristics of the architecture used. When the image correction stage aligns with the feature extraction strategy within the model, accuracy improvement can be consistently achieved. However, if the initial transformation potentially duplicates existing functions within the network, the effect can be the opposite. These findings underscore the importance of ensuring that pre-processing strategies are well-aligned with the internal design of the model to ensure optimal emotion recognition outcomes, particularly in cases of non-frontal faces.

D. PARTIAL OCCLUSION IN THE MOUTH AREA

Based on the results of the average accuracy changes before and after the application of the method on faces with hand occlusion in the mouth area, it is evident that the observed decrease in performance is relatively insignificant across most models. For instance, the CNN model only experienced a very slight increase (+0.25%), while DeepFace and FaceLib showed average decreases (-2.93% and -2.92%, respectively). SVM also exhibited a slight decline (-0.85%). It can be concluded that the proposed method did not have a significant impact on improving emotion recognition performance in the case of hand occlusion in the mouth area, and did not produce substantial differences compared to the initial condition. This indicates that the characteristic features of expressions in the mouth area, which are covered by the hand, are difficult to normalize with the approach used. As a result, the model did not gain any substantial new information for the classification process.

These results align with the phenomenon observed in facial expression recognition research when the mouth area is occluded, such as in the case of wearing medical masks. Various studies have shown that emotions such as happiness, sadness, and anger are more difficult to recognize when the mouth and nose are covered by a mask [51][52]. The decrease in accuracy can also be attributed to the loss of several important expression actions in that area, such as smiling, lip movements, or nose wrinkles, all of which contribute significantly to detecting specific emotions. In these conditions, the system tends to rely more on features from the upper face, such as the eyes and eyebrows, which are less informative for certain emotions. Moreover, this dependency often results in ambiguous emotion estimation, particularly between expressions with similar upper-face cues, such as anger and disgust. From a psychological

perspective, the mouth region provides strong and distinctive cues that humans also use to interpret emotional intent, and its absence not only challenges machine models but also human perception accuracy. Consequently, many recent studies have focused on enhancing feature extraction from the visible facial regions, integrating contextual information, or employing data augmentation techniques to compensate for the missing lower-face patterns and restore recognition performance under partial occlusion.

E. STUDENT ENGAGEMENT DURING ONLINE LEARNING

The results of this study indicate that integrating the proposed method with the FER system in an asynchronous learning scenario can provide a quantitative depiction of students' engagement patterns while watching instructional videos. The system not only identifies emotion percentages but also converts them into more applicable engagement statuses, such as highly engaged, confused, bored, sleepy, and very not engaged. Through this approach, the evaluation of student engagement becomes more measurable and objective, enabling educators to gain deeper insight into students' affective responses toward the learning material. The presence of engagement-insight visualizations further strengthens the analytical function, as educators can directly observe engagement distribution throughout the duration of the video.

The evaluation conducted on three students with different scripted behaviors demonstrates the system's capability to distinguish various engagement patterns. For example, Student 1 showed a dominance of boredom at 53.33%, Student 2 was dominated by the confused category at 51.11%, whereas Student 3 exhibited the highest level of highly engaged at 51.11%. These differences indicate that the system is able to consistently capture variations in affective expressions and behaviors, confirming the model's sensitivity to relevant emotional features. Thus, the proposed method is not only effective in detecting facial emotions but also contributes to a more comprehensive understanding of students' engagement quality in online learning.

F. FUTURE WORKS

Based on the results of the conducted research, several key areas have been identified that require further development to improve the robustness of the Facial Expression Recognition (FER) system under partial occlusion conditions. Although the proposed method provides an initial contribution in addressing hand-occluded faces, the limitations encountered, such as inconsistent effectiveness on deep learning-based models, hand detection failures in certain conditions, and minimal improvement in mouth area occlusion, indicate the need for a more adaptive and comprehensive approach. These challenges align with findings from other studies, which emphasize that facial occlusion, whether caused by masks, accessories, or hands, remains a significant barrier in computer vision-based FER systems, particularly when key facial expression features, such as the eyes, nose, and mouth, are covered.

Therefore, the development of solutions capable of addressing a wide range of occlusions, improving detection accuracy in extreme poses, and integrating multi-modal information should be a top priority for future research. These advancements are necessary to create a FER system that is truly robust in real-world applications. Several potential developments for making FER systems more resilient to various conditions are outlined below, focusing on areas such as adaptive pre-processing techniques, enhanced pose recognition models, and leveraging additional data sources like audio or physiological signals.

VII. CONCLUSION

This study shows that the proposed method yields the strongest results in cases of faces partially occluded vertically by hands, where key facial areas such as the eyes, cheeks, and eyebrows are covered. In this condition, several models, including CNN and DeepFace, exhibited clear performance improvements, with an average accuracy increase of 33.79% for CNN and 14.29% for DeepFace. Meanwhile, for faces with pose variations (non-frontal), the method also resulted in an accuracy increase, although the improvement was not significant, indicating the model's potential to adapt to different face orientations. Conversely, for faces with occlusion in the mouth area, the method did not have a meaningful impact on emotion detection accuracy, with some models, such as DeepFace (-2.93%) and FaceLib (-2.92%), actually experiencing a performance decrease. This indicates that the loss of information in the mouth area significantly affects the system's ability to interpret emotional expressions, especially for expressions like happy, sad, and fear, which generally rely heavily on lip movements.

Nevertheless, the results of this study still demonstrate that the proposed approach has the potential to enhance the robustness of expression recognition systems under certain partial occlusion conditions. Other factors, such as the failure of the MediaPipe hand detector to detect the presence of hands, also affect the overall system performance, particularly in low lighting conditions or extreme hand poses. This limitation opens the door for the development of more occlusion-adaptive models that are more sensitive to variations in hand position in the future.

It is important to note that the findings of this study are limited to the dataset used in this experiment, which has specific characteristics and image distributions. Therefore, the performance of the method may vary when tested on datasets with more diverse resolutions, more complex occlusion levels, or different visual backgrounds. Further research using multi-scenario datasets and multi-modality approaches is recommended to confirm the effectiveness of the method more broadly and expand its application to real-world scenarios.

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FADHIL HIDAYAT (Member, IEEE) was born on September 25, 1986, in Payakumbuh. He completed his master's degree at Institut Teknologi Bandung in 2011, and Doctor's degree at Institut Teknologi Bandung in 2018. He is a lecturer in Institut Teknologi Bandung and Joined Smart City & Community Innovation Center.



KUSPRASAPTA MUTIJARSA is currently an Assistant Professor with the School of Electrical Engineering and Informatics, Bandung Institute of Technology, Indonesia. He is also the Head of the Autonomous Vehicle Research Group. His current research interests include autonomous and intelligent systems, artificial intelligence and machine learning, robotics, and autonomous vehicles.



ARI APRIANSYAH obtained his bachelor degree in Computer System from Universitas Tanjungpura in Pontianak in 2016. Currently, he is a Master's student in Smart-X at STEI Bandung Institute of Technology, a recipient of the Smart-X scholarship from the Ministry of Communication and Informatics. His research interests primarily focus on computer vision and embedded systems.



ARRY AKHMAD ARMAN is currently an Associate Professor with the School of Electrical Engineering and Informatics, Bandung Institute of Technology (ITB), Indonesia. His main research interests include IT governance, enterprise architecture, smart cities governance, and speech recognition.



I GUSTI BAGUS BASKARA NUGRAHA received the B.E. and Master of Science degrees in electrical engineering from the Bandung Institute of Technology (ITB), Indonesia, in 1999 and 2001, respectively, and the Ph.D. degree from The University of Electro Communications, Tokyo, Japan, in 2006. He is currently an Associate Professor with the School of Electrical Engineering and Informatics, ITB. His research interests include management information systems, and cloud and edge computing.



SUHONO HARSO SUPANGKAT received the B.E. degree in electrical engineering from the Bandung Institute of Technology (ITB), Indonesia, in 1986, the Master of Science degree from Meisei University, Japan, in 1994, and the Ph.D. degree from The University of Electro Communications, Tokyo, Japan, in 1998. He was the Head of the Smart City and Community Innovation Center. He is currently a Professor with the School of Electrical Engineering and Informatics, ITB. His research interests include smart cities concepts, smart city governance, and the Internet of Things.